MODEL QUESTION PAPER

REVISION-2010 Reg. No…………………

SUB CODE:3077 Signature……………….

THIRD SEMESTER DIPLOMA EXAMINATION IN ENGINEERING /TECHNOLOGY-OCTOBER 2011

ELECTRICAL & ELECTRONIC INSTRUMENTS

(Common for EI & IT)

Time :3hrs. Maximum Marks:100

PART- A

(Answer the question in one or two sentences .Each question carries 2 marks)

I 1. Define Deflection sensitivity

2. What you mean by slewing speed in XY recorder

3. What is the function of time base circuit in CRO

4.Find the value of shunt resistance required for using a µA meter movement with an internal resistance of 250Ω for measuring 0-500mA

5.Give any two errors in Dynamometer type wattmeter

(5X2=10)

PART- B

(Answer any five questions. Each question carries 6marks)

II 1.With the help of neat diagram ,convert the basic D'Arsonval movement into a dc voltmeter

2.The screen of CRO shows a sinusoidal signal of total vertical occupancy of 3cm and total horizontal occupancy of 2cm. The volt/div& Time/div adjustments are on 2v/div &2msec/div respectively. Calculate the r.m.s value of voltage and its frequency

3.Explain with neat circuit the theory and working of DC Wheatstone bridge

4.Explain the Electrostatic deflection and derive the expression for deflection sensitivity of CRO

5.Describe about fluid friction damping

6.With the help of neat sketch explain the measurement of voltage &current using digital multimeter

7.Describe with block schematic ,the operation of logic analyzer

(5x6=30)

PART –C

(Answer one full question from each unit. Each question carries 15 marks)

UNIT –I

III 1.With the help of neat diagram explain the working of attraction type moving iron instrument (8)

2.Describe the methods of measuring ac voltage and current using analog mulitmeter (7)

OR

IV 1. With neat diagram explain the working of air friction damping
(8)

2.Draw &explain the DC voltage &current measuring circuit of digital multimeter
(7)

UNIT –II

V 1.Describe with suitable diagram the principle of dynamometer type wattmeter (8)

2.Explain the working principle of Q-meter
(7)

OR

VI 1.Explain with a circuit the principle of impedance measurement using Maxwell's Bridge (8)

2 Describe with diagram the construction of single phase energymeter (7)

UNIT –III

VII 1.Explain the Block diagram of CRO
(8)

2.Explain the measurement of frequency using CRO
(7)

OR

VIII 1.Explain with the schematic the principle of operation of pulse generartor (8)

2.Explain with schematic the principle of operation of digital storage Oscilloscope (7)

UNIT –IV

IX 1. .Explain with the Block diagram the principle of measurerment of time &frequency using universal timer /counter
(8)

2. .Explain with diagram the working of circular chart data recorder
(7)

OR

X 1.Describe the block diagram and principle of spectrum analyzer
(8)

2.Explain with the schematic the working of galvanometric recorder
(7)
